

EE309 POWER SYSTEMS TERM PROJECT

Detection of Energy Theft and Defective Smart Meters in Smart Grids Using Linear Regression

PRESENTED BY:-

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INTRODUCTION

- Pragy Bagoriya

Background

- Electricity distribution systems suffer from different types of losses
- Losses are divided into:
 - Technical losses**
 - Non-technical losses (NTL)**
- Energy theft is a major form of non-technical loss

What is Energy Theft?

- Illegal consumption of electricity without proper billing
- Common causes:
 - Meter tampering
 - Bypassing the meter
 - Illegal connections
 - False meter readings

Why is Energy Theft a Serious Problem?

- Causes huge financial loss to utility companies
- Reduces grid reliability
- Affects power quality and system planning
- Increases burden on honest consumers

Smart Grids and Smart Meters

- Traditional grids are being replaced by **smart grids**
- Smart grids use **smart meters**
- Smart meters send consumption data automatically
- Part of **Advanced Metering Infrastructure (AMI)**

New Challenges in Smart Grids

- Smart grids improve monitoring
- But they also create new risks:
 - Cyberattacks
 - Data manipulation
 - Communication attacks
 - Faulty smart meter readings

SOLUTION

THE TWO METHODOLOGIES

Methodology 1: Linear Regression-Based Detection of Abnormal Smart Meters (LR-ETDM)

- Uses electricity consumption data from smart meters and collector meter
- Applies linear regression to model normal consumption behavior
- Detects abnormal users through anomaly coefficients
- Helps identify suspicious consumers

Methodology 2: Detection of Defective Meters with Time/Location Identification (CVLR-ETDM)

- Extends the first method
- Uses additional detection variables
- Detects defective smart meters
- Helps identify when and where abnormality occurs

METHODOLOGY-1

- Muskan Garg

1. Energy Discrepancy

Total Supply = Sum of all Consumer Usage

$$c_{ti} = \sum_{n=1}^N p_{ti,n} + k + h + c$$

Where,

c_{ti} - total electricity supplied

$p_{ti,n}$ - electricity reported by consumer n

k- technical losses

h- energy theft

c- faulty smart meter error

2. Discrepancy Equation

$$y_{ti} = c_{ti} - \sum_{n=1}^N p_{ti,n}$$

where,

y_{ti} - discrepancy in readings

c_{ti} - energy supplied

$\sum_{n=1}^N p_{ti,n}$ - total sum of energy consumed by N consumers

Value	Meaning
$y_{ti} > 0$	Energy theft
$y_{ti} < 0$	Meter error / abnormal behavior
$y_{ti} = 0$	System is normal

3. Formulation of Linear System of Equation

$$a_{1pti,1} + a_{2pti,2} + \dots + a_{npti,n} = y_{ti}$$

These equations are written for multiple time intervals.

converted into matrix form: **$Pa=y$**

where,

P = consumer energy data matrix

a = anomaly coefficient vector- Each consumer is assigned a coefficient that represents abnormal behavior.

y = discrepancy vector

Value	Meaning
a=0	Honest Consumer
a>0	Energy Theft
a<0	Faulty Smart meter

$$(a_1+1)p_{ti,1} + (a_2+1)p_{ti,2} + \dots + (a_n+1)p_{ti,n} = c_{ti}$$

Here, **instead of a_1 , we will use (a_1+1)**

Because, when **$a_n=0$, reading remains unchanged.**

Thus, using this we get-

$$a_{1pti,1} + a_{2pti,2} + \dots + a_{npti,n} = c_{ti} - \sum_{n=1}^N p_{ti,n}$$

Note- The model **does not include technical losses (TLs)** like transmission losses in power lines or transformers.

However, if TLs are calculated separately, they can be **subtracted from the discrepancy value y** to make the detection more accurate.

4. Solve using Multiple Linear Regression (LR-ETDM)

$$a_{1pti,1} + a_{2pti,2} + \dots + a_{npti,n} = Y_{ti}$$

Now, we have to find the value of anomaly coefficients.

Statistical Model used:

$Y_{ti} \sim N(\mu_i, \sigma^2)$ - Normal Distribution

$d_{ti} = \alpha + \sum_{n=1}^N a_{npti,n}$ - mean of distribution and $\alpha=0$ (response is entirely dependent on predictors)

When $\alpha = 0$, This equation forms a **hyperplane in N-dimensional space**.

Assumptions-

1. The Predictors Are uncorrelated with each other.
2. The Coefficients an never change throughout the period of observation. This assumption only holds true when the consumers cheat consistently throughout the period of observation.

Solving- It has been shown in that the maximum likelihood estimate of the coefficients a are those that minimize the residual sum of squares between y_{ti} and d_{ti} . If P_i of full column rank, then a is given by:

$$a = (P^T P)^{-1} P^T y$$

Hypothesis Test-

H0 : $a_n = 0$ - honest consumer

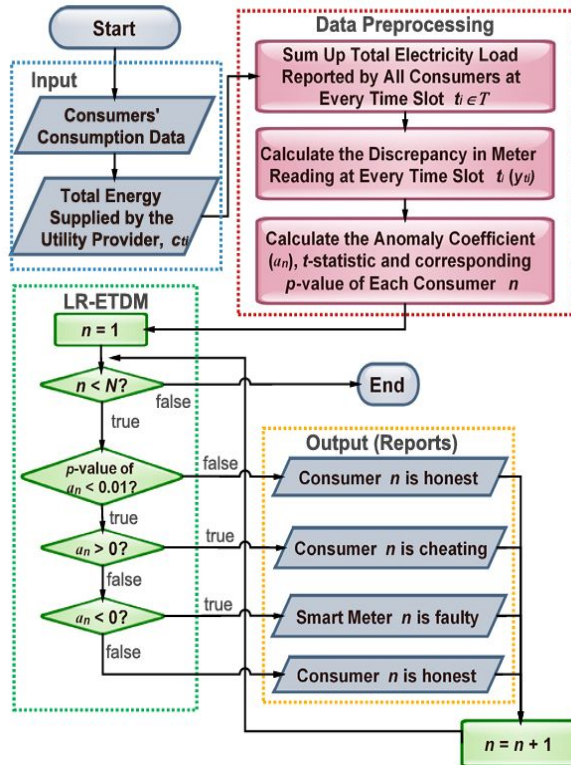
H1 : $a_n \neq 0$ - dishonest consumer

Now, The t-statistic is calculated as: **$t = a_n / SE(a_n)$** , SE is standard error of the coefficient and we get the p-value.

If,

1. **p-value < 0.01**, the null hypothesis is rejected, indicating that the consumer has abnormal behavior (energy theft if $a_n > 0$, faulty meter if $a_n < 0$).
2. **p-value > 0.01**, consumer is honest

5. LR-ETDM Detection Algorithm



Steps-

1. Collect smart meter consumption data.
2. Compute Total Consumer Consumption.
3. Compute Energy Discrepancy y .
4. Construct linear equation system.
5. Estimate Anomaly coefficients.
6. Compute t-statistic and p-values.
7. Check significance.
8. Classify the consumer.
9. Repeat for all consumers.
10. Generate final output.

Limitations-

1. Assume Constant Anomaly Coefficient.
2. Cannot Detect Intermittent Energy Theft.
3. Sensitive to Changes in Consumer Behavior.
4. Does Not Identify the Time of Fraud

6. Results of LR-ETDM Implementation

The graphs show the **estimated anomaly coefficient a_n** for each consumer.

X-axis: Consumer ID

Y-axis: Anomaly coefficient a_n

Interpretation of p and a_n

$p \geq 0.01 \rightarrow$ Honest consumer

$p < 0.01$ and $a_n > 0 \rightarrow$ Energy theft (consumer reports less usage)

$p < 0.01$ and $a_n < 0 \rightarrow$ Faulty smart meter (reports higher consumption)

The t-statistic and p-value verify whether the anomaly coefficient is reliable. The sign of a_n then determines whether the abnormality corresponds to energy theft or meter malfunction.

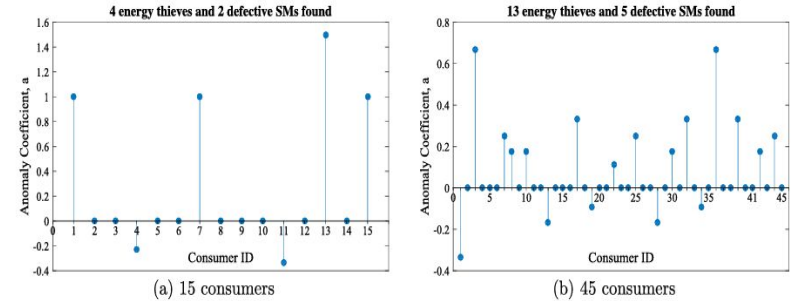


Fig. 4. Value of a obtained by LR-ETDM when a is constant.

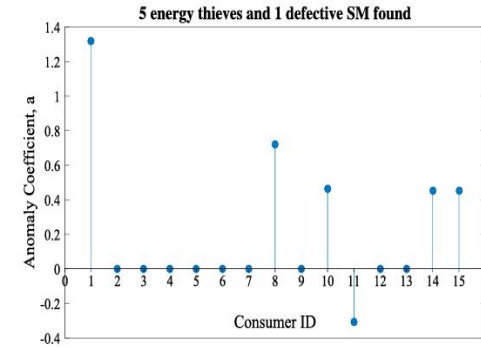


Fig. 5. Value of a obtained by LR-ETDM when a is variable.

METHODOLOGY-2

- Sakshi Sinha

The Failure of LR-ETDM on variable theft

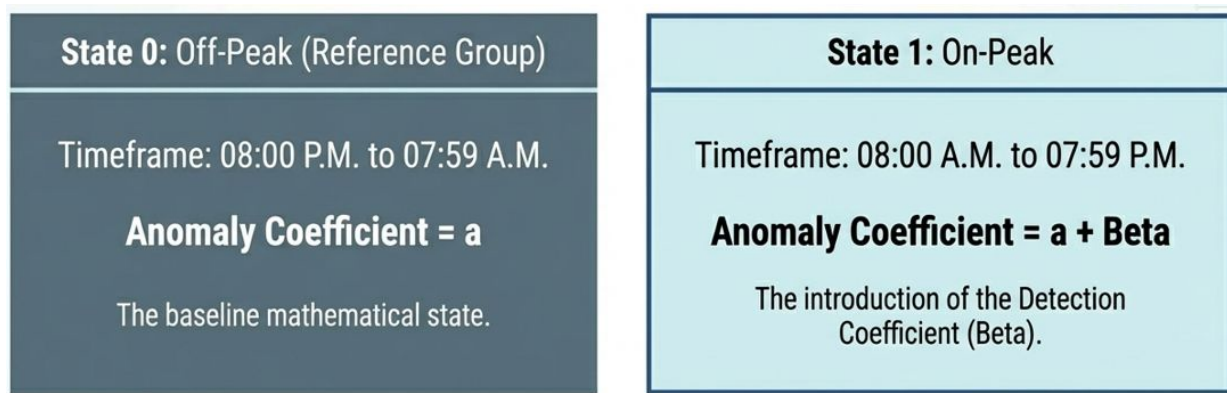


Standard Multiple Linear Regression (LR-ETDM) assumes fraud is a constant state. However, **Time-of-Use (TOU)** pricing incentivises consumers to shift theft exclusively to peak hours. When anomalous behaviour is inconsistent, standard regression fails to detect it.

Solution: CVLR-ETDM

“Categorical Variable-Enhanced Linear Regression based scheme for Energy Theft and Defective Meter Detection”

A categorical variable represents data belonging to **different groups rather than continuous values**. In CVLR-ETDM, the **24-hour cycle is divided into two states**:



To solve the time-variant cheating problem, the **CVLR-ETDM model introduces Categorical Variables (Dummy Coding)**. By splitting the 24-hour cycle into two distinct states, we force the algorithm to evaluate **Beta**. If **Beta** is anything other than zero, it mathematically flags the exact moment a consumer alters their behaviour.

6. Introducing Categorical Variables

A **categorical variable** represents data that belongs to **different categories or groups** rather than continuous values.

Off-peak time =0

On-peak time=1

Dummy code- we include the categorical variables, x_i for $i=1; 2; \dots; N$ to categorize the time of fraud or metering defect of consumers 1; 2;...; N.

0- 8:00 PM – 7:59 AM (on peak)

1- 8:00 AM – 7:59 PM (off peak)

The category which is designated as 0 (i.e., off-peak) in the categorical variable is known as the reference group.

Detection Coefficients

To incorporate categorical variables, the regression model introduces a new coefficient **β (beta)** called the detection coefficient. This coefficient captures **changes in anomaly behaviour during on-peak periods**.

Equation becomes-

$$a_1 p_{t1,1} + \dots + a_N p_{t1,N} + \beta_1 p_{t1,1} x_1 + \dots + \beta_N p_{t1,N} x_N = y_{t1}$$

$$y_t = \sum_{i=1}^N a_i p_{t,i} + \sum_{i=1}^N \beta_i p_{t,i} x_i$$

where,

a_i : anomaly coefficient during off-peak

β_i : detection coefficient indicating change during peak time

x_i : dummy variable (0 = off-peak, 1 = on-peak)

$p_{t,i}$: power consumption of user i at time t

On and Off Peak Regression Model

When $x=0$, Off

$$a_1 p_{to;1} + a_N p_{to;N} + \beta_1 p_{to;1} \cdot 0 + \dots + \beta_N p_{to;N} \cdot 0 = y_{to} ;$$

$$\mathbf{P}^{off} \mathbf{a} = \mathbf{y}^{off}$$

When $x=1$, On

$$a_1 p_{to;1} + a_N p_{to;N} + \beta_1 p_{to;1} \cdot 1 + \dots + \beta_N p_{to;N} \cdot 1 = y_{to} ;$$

$$\mathbf{P}^{on} (\mathbf{a} + \boldsymbol{\beta}) = \mathbf{y}^{on}$$

Thus, relation between these can be given by-

$$\mathbf{P}_{g}^{aug} \rightarrow \left[\begin{array}{c|c} \mathbf{P}^{off} & \mathbf{0} \\ \hline \mathbf{P}^{peak} & \mathbf{P}^{peak} \end{array} \right] \left[\begin{array}{c} \mathbf{a} \\ \boldsymbol{\beta} \end{array} \right] = \left[\begin{array}{c} \mathbf{y}^{off} \\ \mathbf{y}^{peak} \end{array} \right]$$

The maximum likelihood estimator of our regression coefficients are thus computed by:

$$\left[\begin{array}{c} \mathbf{a} \\ \boldsymbol{\beta} \end{array} \right] = ((\mathbf{P}^{aug})' \mathbf{P}^{aug})^{-1} (\mathbf{P}^{aug})' \mathbf{y}$$

Consumer Behaviour

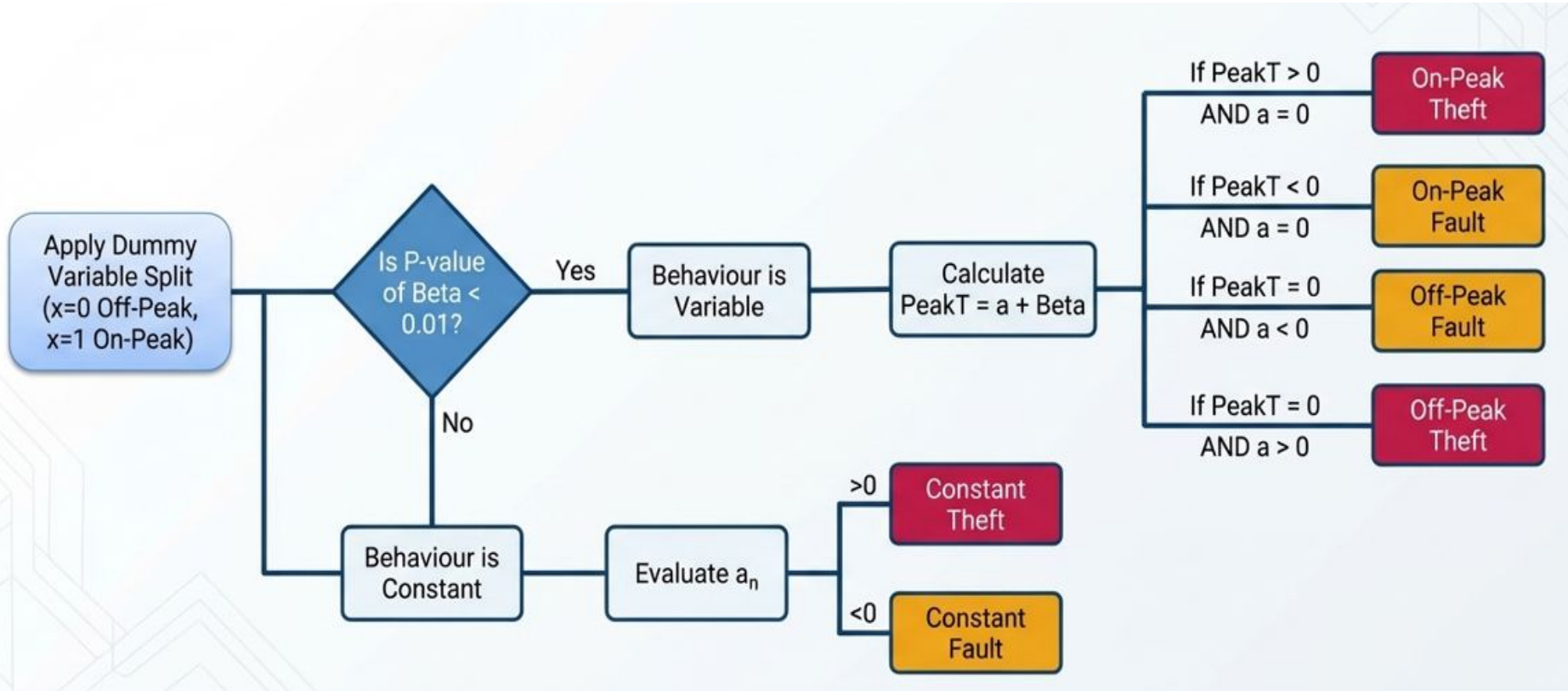
Based on values of a and β , the possible scenarios are:

1. $a=0, \beta=0 \rightarrow$ Honest consumer
2. $a>0, \beta=0 \rightarrow$ Stealing all the time
3. $a<0, \beta=0 \rightarrow$ Faulty smart meter all the time
4. $a=0, \beta>0 \rightarrow$ Theft only during peak hours
5. $a=0, \beta<0 \rightarrow$ Meter faulty during peak hours
6. $a>0, \beta=-a \rightarrow$ Theft only during off-peak
7. $a=-\beta, \beta>0 \rightarrow$ Meter faulty during off-peak

The Diagnostic Truth Table

Off-Peak (a)	Behaviour Shift (Beta)	On-Peak (a + Beta)	System Diagnosis
0	0	0	Honest
> 0	0	> 0	Constant Theft
< 0	0	< 0	Constant Fault
0	> 0	> 0	Theft During On-Peak Only
0	< 0	< 0	Fault During On-Peak Only
> 0	-a	0	Theft During Off-Peak Only
-b	> 0	0	Fault During Off-Peak Only

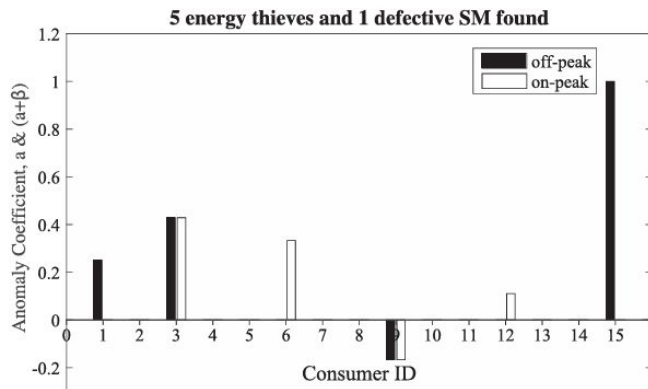
CVLR-ETDM Advanced Decision Logic



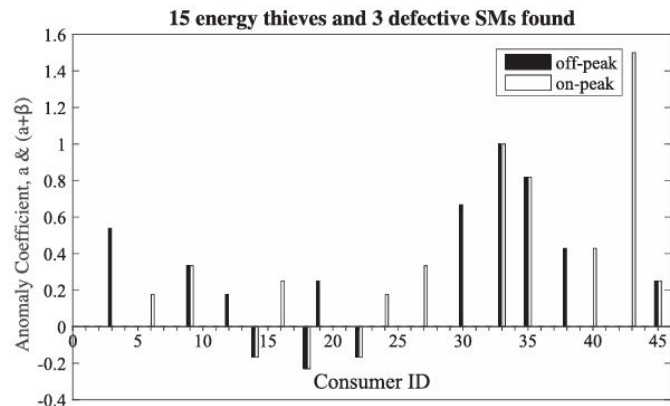
Performance Evaluation of CVLR-ETDM

Dataset & Setup

1. **The Source Data:** Half-hourly smart energy data extracted from the Irish Smart Energy Trial (commercial and residential).
2. **The Injection Rate:** A rigorous 40% anomaly rate is injected into the networks. This means 4 out of 10 meters are deliberately compromised or broken.
3. **The Threat Profiles:** Algorithms are tested against randomly distributed behaviours: constant theft, variable on-peak theft, variable off-peak theft, and constant system failures.



(a) 15 consumers



(b) 45 consumers

Advantages of CVLR-ETDM

Hardware Agnostic & Cost Efficient

Requires zero new hardware, physical sensors, or manual inspections. It operates entirely mathematically on the existing Advanced Metering Infrastructure, dramatically reducing operational expenditure.

Zero Historical Profiling

Unlike Support Vector Machines or Neural Networks, CVLR-ETDM does not require vast databases of historical training data. It flags anomalies based purely on near-time LSE calculations.

Temporal Precision

By incorporating dummy coding, it is the first linear regression model capable of pinning down the exact time-of-day a consumer alters their cheating behaviour, closing the Time-of-Use loophole.

Limitations / Future Work

Vector 1: Noise Tolerance

Current Limitation

The model assumes all non-technical losses derive solely from theft or defective meters. It currently lacks dynamic environmental noise filtering.

Future Work

Integrating dynamic Technical Loss (TL) calculations (e.g., non-ohmic losses from distribution transformers) as a subtractive baseline prior to running the Linear System of Equations.

Vector 2: Privacy Preservation

Current Limitation

The collector requires unencrypted, highly granular half-hourly data from every individual household, posing a significant consumer data privacy risk.

Future Work

Developing cryptographic data-concealment algorithms (e.g., secure multiparty computation) that allow the CVLR-ETDM matrix to solve for anomalies without exposing the raw reporting data of honest citizens.

Conclusion: Securing the Smart Grid

The Threat

Advanced Metering Infrastructure exposes utilities to sophisticated, temporally-variable data tampering that easily evades traditional physical checks and basic regression models.

The Solution

By introducing categorical variables (dummy coding) into Multiple Linear Regression, CVLR-ETDM successfully maps and isolates dynamic consumer behaviour.

The Impact

Utilities can now pinpoint exactly who is stealing energy, when they are stealing it, and which meters are mechanically defective, saving billions in Non-Technical Losses with zero additional hardware.

RESULTS

Methodology of Implementation for LR-ETDM:

FOR each dataset (Dataset 1 and Dataset 2):

LOAD:
Files

EXTRACT:
unique meter IDs $\rightarrow$ N
unique time slots $\rightarrow$ T

BUILD matrix P (T x N):
FOR each meter n:
sort readings by time
store usage values in column n

COMPUTE:
total supplied energy $c(t)$
residual $y(t) = c(t) - \text{sum of all meter readings}$

CREATE regression model:
 $y = a_1 \cdot C_1 + a_2 \cdot C_2 + \dots + a_N \cdot C_N$

Linear regression
coefficients (a_n)
p-values
t-statistics

CLASSIFY each meter:
IF p-value < 0.01:
IF $a_n > 0 \rightarrow \text{label} = \text{"THIEF"}$
ELSE $\rightarrow \text{label} = \text{"FAULTY"}$
ELSE:
label = "Honest"

COMPUTE scaling factor:
 $m = 1 / (1 + a_n)$

COMPARE with ground truth:
count correct detections
compute accuracy

END

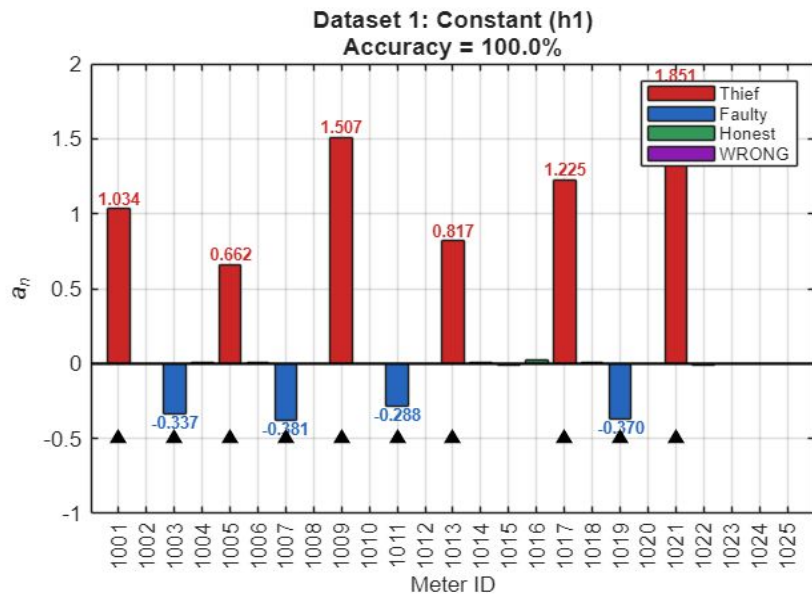
Results of LR-ETDM Implementation for constant theft

According to the paper, LR-ETDM works for the data that has constant theft, and thus the results for the dataset with constant theft is 100% according to the computation.

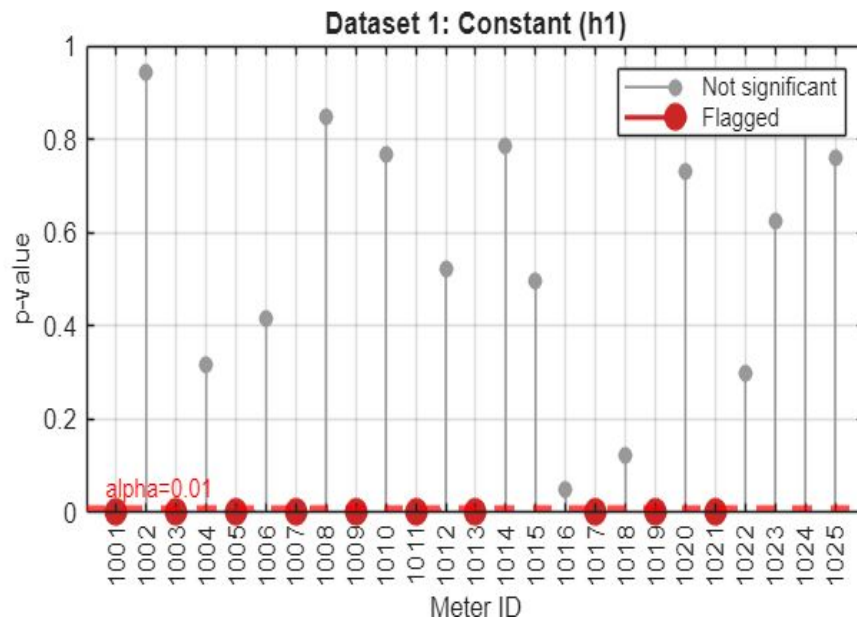
FOR DATASET 1:

LR-ETDM DATASET 1: Constant Theft (h1)							
m<1 = theft (under-reporting) m=1 = honest m>1 = faulty (over-reporting)							
MeterID	a_n	SE(a_n)	t-stat	p-value	m	Detected	Truth
1001	1.0336	0.0204	50.6867	0.0000	0.4917	THIEF	THIEF
1002	0.0005	0.0069	0.0702	0.9446	1.0000	Honest	Honest
1003	-0.3368	0.0054	-61.8014	0.0000	1.5078	FAULTY	FAULTY
1004	0.0109	0.0106	1.0244	0.3163	1.0000	Honest	Honest
1005	0.6616	0.0136	48.6413	0.0000	0.6018	THIEF	THIEF
1006	0.0091	0.0110	0.8260	0.4173	1.0000	Honest	Honest
1007	-0.3814	0.0094	-40.7739	0.0000	1.6166	FAULTY	FAULTY
1008	-0.0012	0.0060	-0.1962	0.8462	1.0000	Honest	Honest
1009	1.5074	0.0178	84.8237	0.0000	0.3988	THIEF	THIEF
1010	-0.0022	0.0072	-0.3016	0.7657	1.0000	Honest	Honest
1011	-0.2878	0.0055	-52.1095	0.0000	1.4040	FAULTY	FAULTY
1012	-0.0039	0.0060	-0.6484	0.5232	1.0000	Honest	Honest
1013	0.8174	0.0108	75.6192	0.0000	0.5502	THIEF	THIEF
1014	0.0030	0.0111	0.2740	0.7865	1.0000	Honest	Honest
1015	-0.0078	0.0113	-0.6922	0.4958	1.0000	Honest	Honest
1016	0.0242	0.0116	2.0796	0.0489	1.0000	Honest	Honest
1017	1.2249	0.0279	43.9006	0.0000	0.4495	THIEF	THIEF
1018	0.0107	0.0067	1.5964	0.1241	1.0000	Honest	Honest
1019	-0.3703	0.0058	-64.1436	0.0000	1.5881	FAULTY	FAULTY
1020	-0.0034	0.0098	-0.3472	0.7316	1.0000	Honest	Honest
1021	1.8509	0.0254	72.7613	0.0000	0.3508	THIEF	THIEF
1022	-0.0119	0.0111	-1.0641	0.2983	1.0000	Honest	Honest
1023	-0.0042	0.0085	-0.4950	0.6253	1.0000	Honest	Honest
1024	-0.0009	0.0097	-0.0910	0.9283	1.0000	Honest	Honest
1025	0.0020	0.0067	0.3069	0.7617	1.0000	Honest	Honest
Accuracy = 100.0% (25 / 25 correct)							

100% accuracy



Value of an obtained by LR-ETDM when an is variable



Results of LR-ETDM Implementation for mixed theft

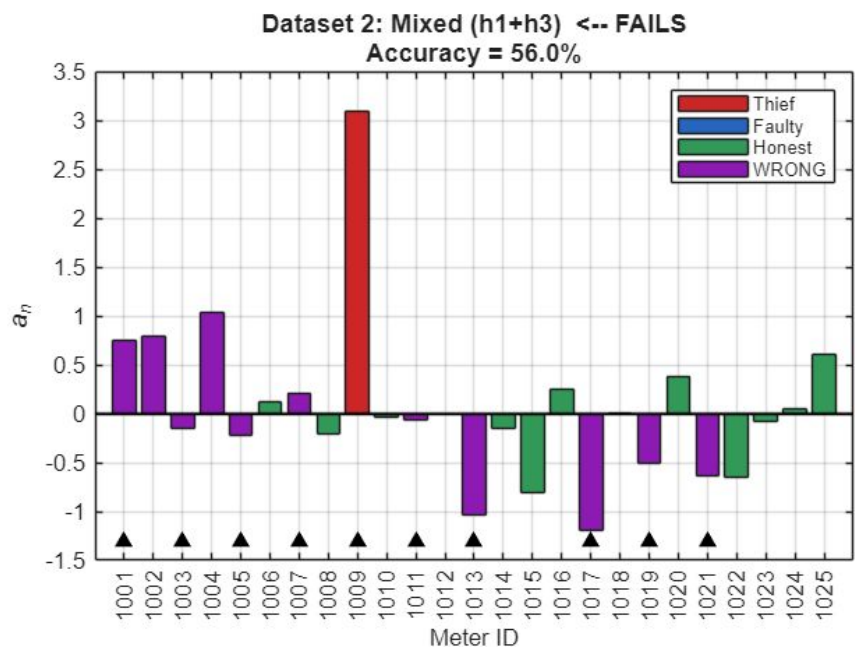
LR-ETDM fails for the data that has mixed theft, and thus the results for the dataset with mixed theft is decreased to 56%, and to counter that we will use CR-ETDM.

FOR DATASET 2:

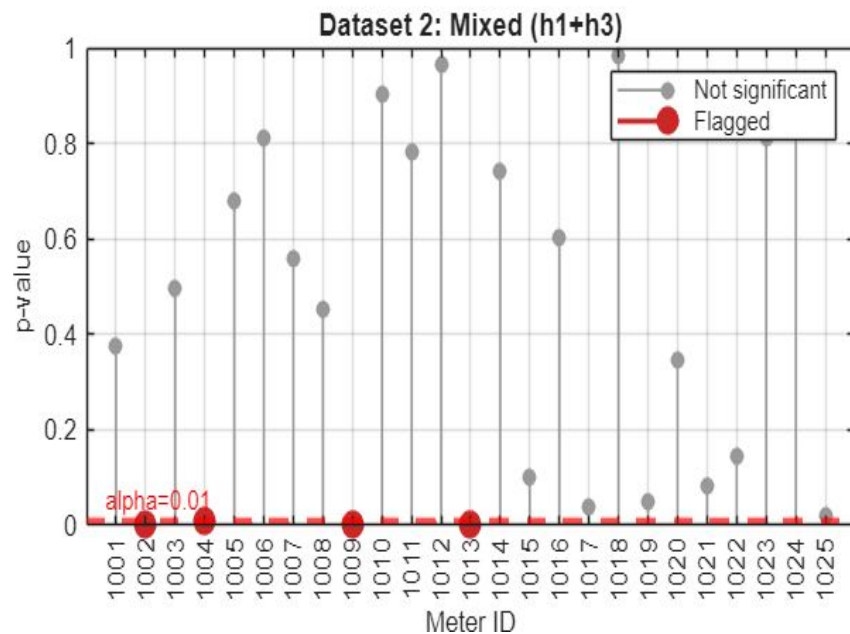
LR-ETDM DATASET 2: Mixed Theft (h1 + h3) <-- WILL FAIL on h3						
m<1 = theft (under-reporting)			m=1 = honest		m>1 = faulty (over-reporting)	
MeterID	a_n	SE(a_n)	t-stat	p-value	m	Detected
Truth						
1001	0.7500	0.8288	0.9049	0.3749	1.0000	Honest
1002	0.7948	0.2335	3.4034	0.0024	0.5572	THIEF
1003	-0.1557	0.2247	-0.6928	0.4953	1.0000	Honest
1004	1.0426	0.3671	2.8405	0.0093	0.4896	THIEF
1005	-0.2226	0.5308	-0.4195	0.6788	1.0000	Honest
1006	0.1185	0.4923	0.2408	0.8119	1.0000	Honest
1007	0.2133	0.3590	0.5942	0.5581	1.0000	Honest
1008	-0.2073	0.2705	-0.7664	0.4513	1.0000	Honest
1009	3.0990	0.6415	4.8312	0.0001	0.2440	THIEF
1010	-0.0324	0.2579	-0.1256	0.9011	1.0000	Honest
1011	-0.0634	0.2267	-0.2797	0.7822	1.0000	Honest
1012	-0.0116	0.2622	-0.0443	0.9650	1.0000	Honest
1013	-1.0406	0.1751	-5.9434	0.0000	-24.6238	FAULTY
1014	-0.1551	0.4652	-0.3333	0.7419	1.0000	Honest
1015	-0.8014	0.4675	-1.7143	0.0999	1.0000	Honest
1016	0.2528	0.4802	0.5265	0.6036	1.0000	Honest
1017	-1.1904	0.5409	-2.2010	0.0380	1.0000	Honest
1018	0.0058	0.2908	0.0201	0.9841	1.0000	Honest
1019	-0.5045	0.2433	-2.0737	0.0495	1.0000	Honest
1020	0.3855	0.4001	0.9635	0.3453	1.0000	Honest
1021	-0.6375	0.3526	-1.8079	0.0837	1.0000	Honest
1022	-0.6553	0.4354	-1.5051	0.1459	1.0000	Honest
1023	-0.0832	0.3476	-0.2393	0.8130	1.0000	Honest
1024	0.0519	0.3736	0.1389	0.8907	1.0000	Honest
1025	0.6116	0.2449	2.4973	0.0201	1.0000	Honest

56% accuracy

Accuracy = 56.0% (14 / 25 correct)



Value of a_n obtained by LR-ETDM when a_n is variable



Methodology of Implementation for CVLR-ETDM:

ALGORITHM: Detect Smart Grid Anomalies (Simplified CVLR-ETDM)

1. GATHER DATA

- Get the total energy supplied to the neighborhood.
- Get the reported energy used by each individual smart meter.
- Calculate the "Missing Energy" (Total Supplied - Sum of Individual Meters).

2. SEPARATE TIME PERIODS

- Split the daily data into two buckets: "Peak Hours" and "Off-Peak Hours".

3. CALCULATE ANOMALY SCORES

- For every consumer, use the "Missing Energy" to calculate two scores:
 - A. Base Score (How they behave during Off-Peak).
 - B. Peak Score (How their behavior changes during Peak hours).

4. FILTER OUT NORMAL NOISE

- Check if the scores are statistically significant. If the scores are tiny or mathematically uncertain, treat them as zero.

5. CLASSIFY EACH CONSUMER

- FOR each consumer:

IF both scores are basically zero:

Label as -> HONEST

ELSE IF their scores show they are reporting LESS energy than they use (Score > 0):

Label as -> THIEF

ELSE IF their scores show they are reporting MORE energy than they use (Score < 0):

Label as -> FAULTY METER

ELSE:

Add label "Constant (All Day)"

6. FINALIZE & REPORT

- Check our labels against the known "Truth" to calculate our final accuracy percentage.
- Save the list of Thieves, Faulty Meters, and Honest consumers.
- Draw graphs to visually show who is stealing or broken during Peak and Off-Peak times.

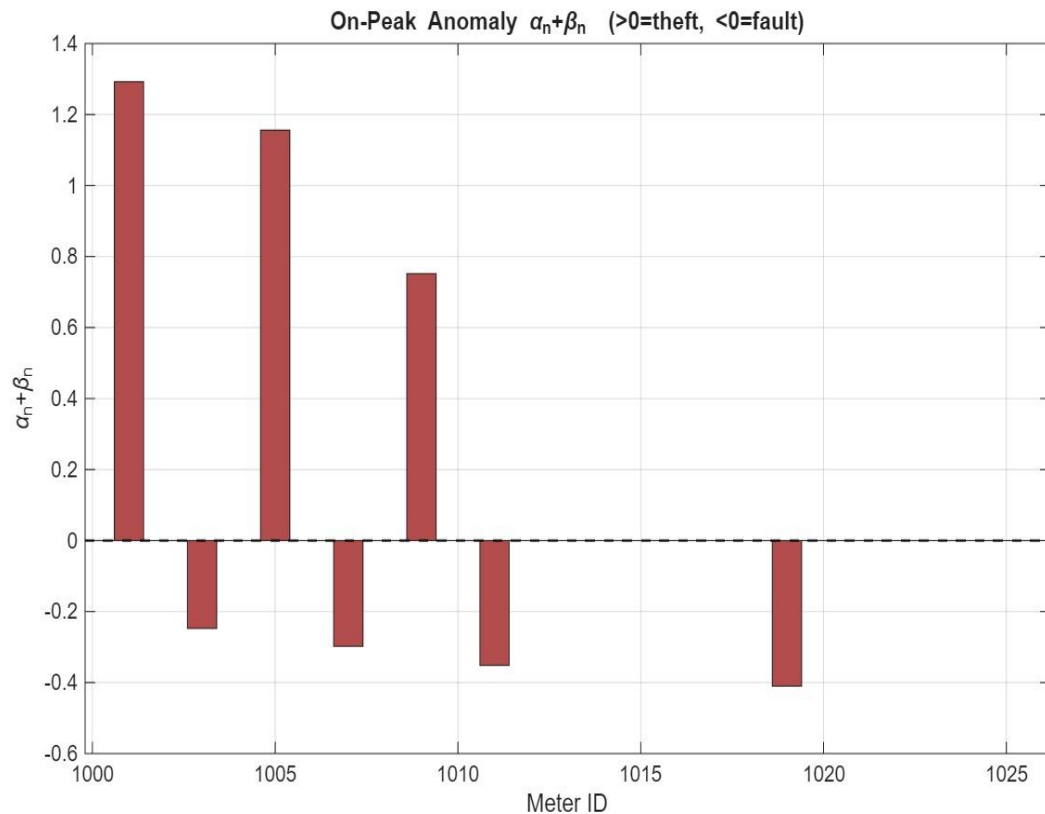
RESULTS OF CVLR-ETDM

[Load] N=25 consumers, T=96 time slots

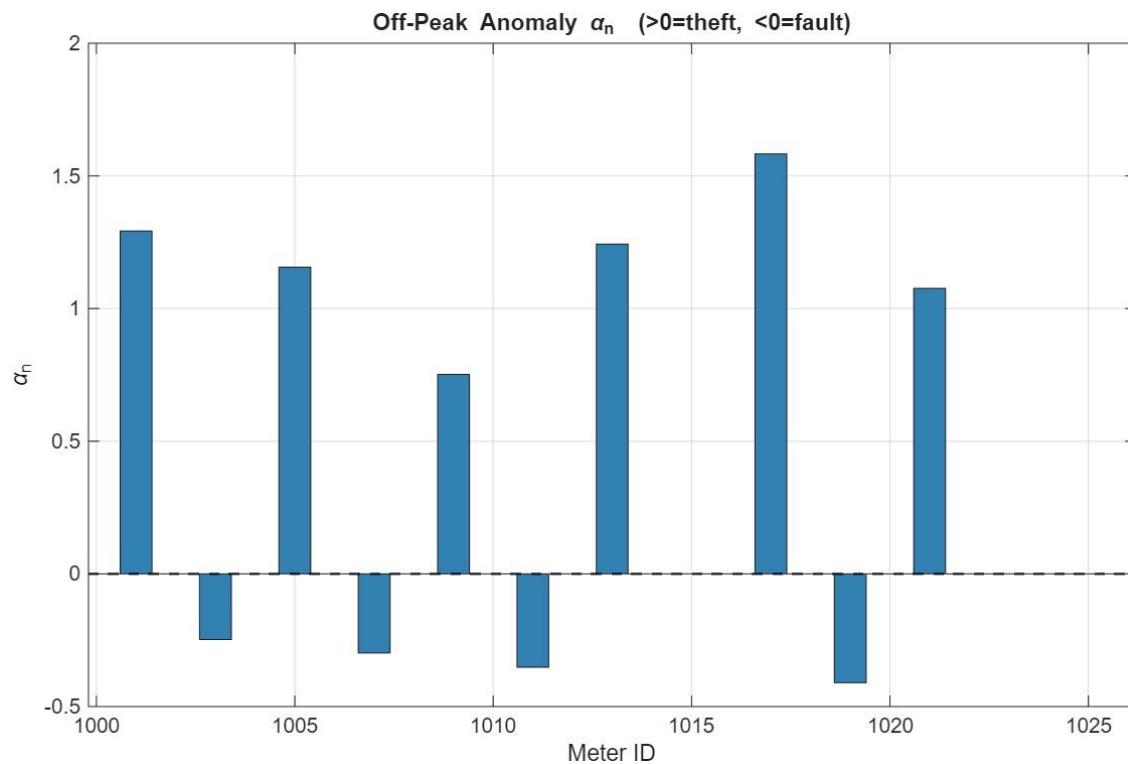
MeterID	Predicted	True_GT	Match
1001	THIEF	THIEF	✓
1002	HONEST	HONEST	✓
1003	FAULTY	FAULTY	✓
1004	HONEST	HONEST	✓
1005	THIEF	THIEF	✓
1006	HONEST	HONEST	✓
1007	FAULTY	FAULTY	✓
1008	HONEST	HONEST	✓
1009	THIEF	THIEF	✓
1010	HONEST	HONEST	✓
1011	FAULTY	FAULTY	✓
1012	THIEF	HONEST	X (GT=HONEST)
1013	THIEF	THIEF	✓
1014	HONEST	HONEST	✓
1015	HONEST	HONEST	✓
1016	HONEST	HONEST	✓
1017	THIEF	THIEF	✓
1018	HONEST	HONEST	✓
1019	FAULTY	FAULTY	✓
1020	HONEST	HONEST	✓
1021	THIEF	THIEF	✓
1022	HONEST	HONEST	✓
1023	HONEST	HONEST	✓
1024	HONEST	HONEST	✓
1025	HONEST	HONEST	✓

Accuracy vs GT: 24/25 = 96.0%

Validation of the CVLR-ETDM model results, achieving a 96.0% accuracy rate across 25 consumers and 96 time slots by successfully matching predicted classifications with Ground Truth (GT).



On-peak anomaly coefficients ($\alpha_n + \beta_n$) calculated by the CVLR-ETDM model, distinguishing between theft (positive values) and faulty meters (negative values) during high-demand periods.



Off-peak anomaly coefficients (α_n) generated by the CVLR-ETDM model, indicating base behavior for consumers where values > 0 represent theft and values < 0 identify faulty smart meters.

MODIFICATIONS

1. From ternary to Multi-Class Detection with Severity Grading

Previously, the system used a ternary classification (Honest, Thief or Faulty) without capturing the type or severity of anomalies. Now, meters are categorized into HONEST, THIEF, FAULTY, or INSPECT with severity levels (MILD, MODERATE, SEVERE), depending on how far the anomaly coefficients deviates from their threshold values.

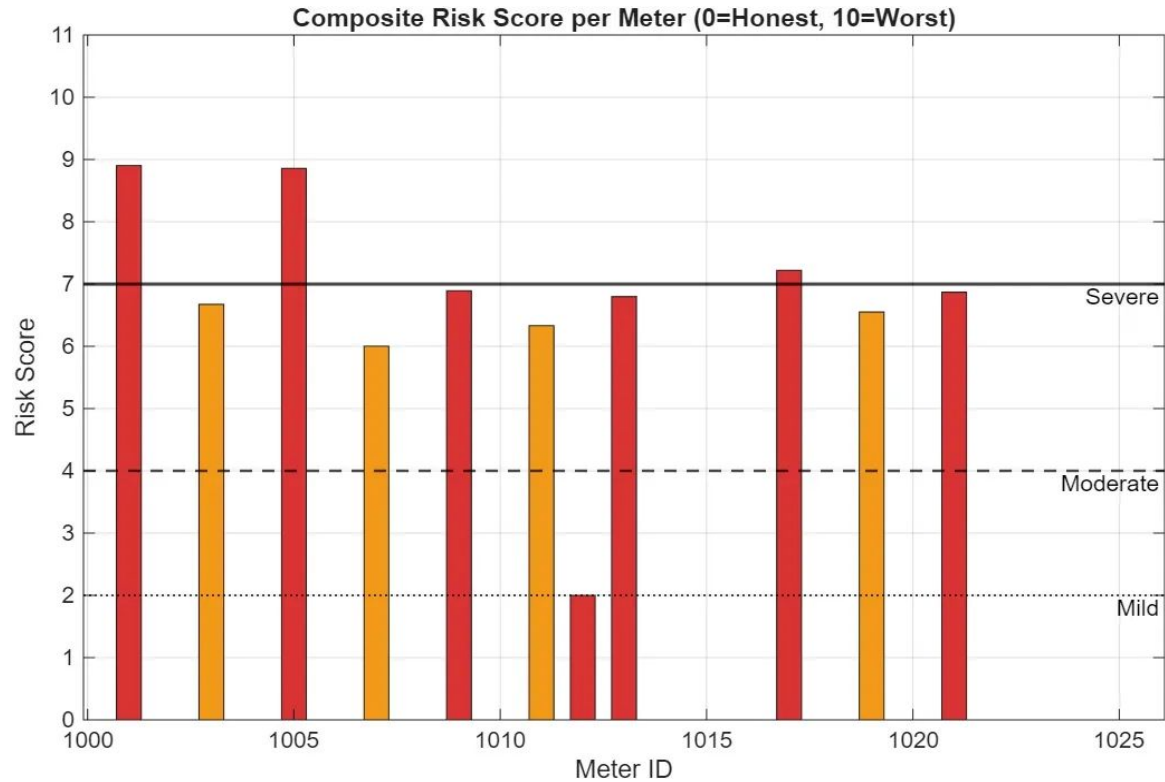
2. From Fixed 2-Period (Peak/Off-Peak) to Adaptive 2–6 Period Time Analysis

Previously, the day was divided into only two fixed periods (peak/off-peak), which could mask time-specific anomalies. Now, the system adaptively splits the day into 2–6 periods based on data availability, enabling more granular and statistically reliable detection.

Modification 1

PRIORITY REPORT				
MeterID	Label	Risk	Loss(kWh)	Action
1001	THIEF_SEVERE_CONSTANT	8.9	34.144	URGENT – schedule on-site inspection within 24 h. Meter under-reports by ~56% all day.
1005	THIEF_SEVERE_CONSTANT	8.9	32.721	URGENT – schedule on-site inspection within 24 h. Meter under-reports by ~54% all day.
1017	THIEF_SEVERE_OFF_PEAK	7.2	37.716	URGENT – schedule on-site inspection within 24 h. Meter under-reports by ~61% during off-peak hours only.
1009	THIEF_MODERATE_MIXED	6.9	27.617	HIGH – schedule inspection within 1 week. Variable under-reporting: ~43% off-peak, ~43% on-peak.
1021	THIEF_SEVERE_OFF_PEAK	6.9	26.925	URGENT – schedule on-site inspection within 24 h. Meter under-reports by ~52% during off-peak hours only.
1013	THIEF_SEVERE_OFF_PEAK	6.8	24.720	URGENT – schedule on-site inspection within 24 h. Meter under-reports by ~55% during off-peak hours only.
1003	FAULTY_MODERATE_MIXED	6.7	20.842	HIGH – meter replacement within 1 week. Variable over-read: ~33% off-peak, ~33% on-peak.
1019	FAULTY_MODERATE_CONSTANT	6.6	41.812	HIGH – meter replacement within 1 week. Meter over-reads by ~70% all day.
1011	FAULTY_MODERATE_CONSTANT	6.3	35.012	HIGH – meter replacement within 1 week. Meter over-reads by ~54% all day.
1007	FAULTY_MODERATE_CONSTANT	6.0	24.807	HIGH – meter replacement within 1 week. Meter over-reads by ~42% all day.
1012	THIEF_MILD_OFF_PEAK	2.0	0.000	MONITOR – flag for next routine inspection cycle. Meter under-reports by ~0% during off-peak hours only.

Priority Report shows that out of 25 meters, 11 were flagged as anomalous. The system not only named them but ranked them by risk score and generated specific action recommendations.



Severe cases (above 7) are clearly distinguished from moderate cases (4–7) and mild cases (below 2) with red bars for thieves and orange for faulty meters, making it immediately clear both the type and urgency of each anomaly.

Modification 2

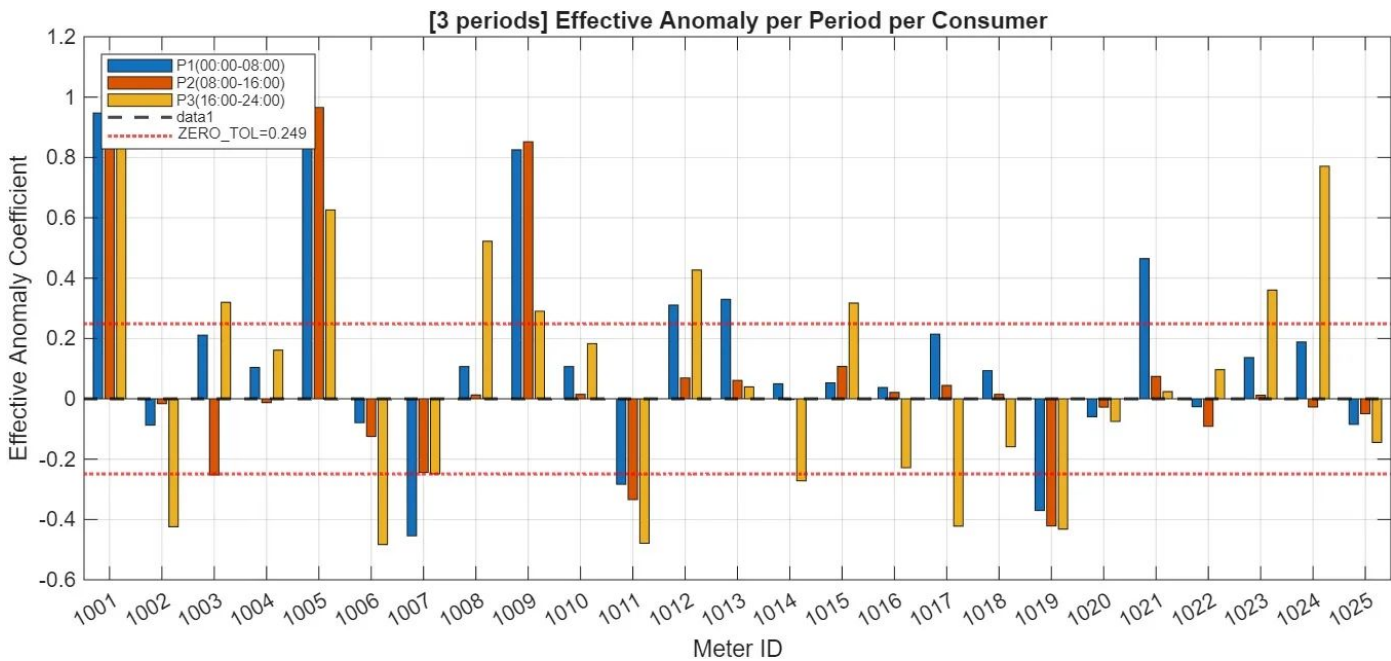
MeterID	Full_Label	Detected	True_GT	Match
1001	THIEF_SEVERE_CONSTANT	THIEF	THIEF	✓
1002	FAULTY_MODERATE_PERIOD_3_ONLY	FAULTY	HONEST	X (GT=HONEST)
1003	INSPECT_MIXED_PERIODS_2_3	INSPECT	FAULTY	X (GT=FAULTY)
1004	HONEST	HONEST	HONEST	✓
1005	THIEF_MODERATE_CONSTANT	THIEF	THIEF	✓
1006	FAULTY_MODERATE_PERIOD_3_ONLY	FAULTY	HONEST	X (GT=HONEST)
1007	FAULTY_MODERATE_PERIOD_1_ONLY	FAULTY	FAULTY	✓
1008	THIEF_MODERATE_PERIOD_3_ONLY	THIEF	HONEST	X (GT=HONEST)
1009	THIEF_MODERATE_ALL_PERIODS_VARIABLE	THIEF	THIEF	✓
1010	HONEST	HONEST	HONEST	✓
1011	FAULTY_MODERATE_CONSTANT	FAULTY	FAULTY	✓
1012	THIEF_MODERATE_PERIODS_1_3	THIEF	HONEST	X (GT=HONEST)
1013	THIEF_MODERATE_PERIOD_1_ONLY	THIEF	THIEF	✓
1014	FAULTY_MODERATE_PERIOD_3_ONLY	FAULTY	HONEST	X (GT=HONEST)
1015	THIEF_MODERATE_PERIOD_3_ONLY	THIEF	HONEST	X (GT=HONEST)
1016	HONEST	HONEST	HONEST	✓
1017	FAULTY_MODERATE_PERIOD_3_ONLY	FAULTY	THIEF	X (GT=THIEF)
1018	HONEST	HONEST	HONEST	✓
1019	FAULTY_MODERATE_CONSTANT	FAULTY	FAULTY	✓
1020	HONEST	HONEST	HONEST	✓
1021	THIEF_MODERATE_PERIOD_1_ONLY	THIEF	THIEF	✓
1022	HONEST	HONEST	HONEST	✓
1023	THIEF_MODERATE_PERIOD_3_ONLY	THIEF	HONEST	X (GT=HONEST)
1024	THIEF_MODERATE_PERIOD_3_ONLY	THIEF	HONEST	X (GT=HONEST)
1025	HONEST	HONEST	HONEST	✓

Base-class accuracy: 15/25 = 60.0%

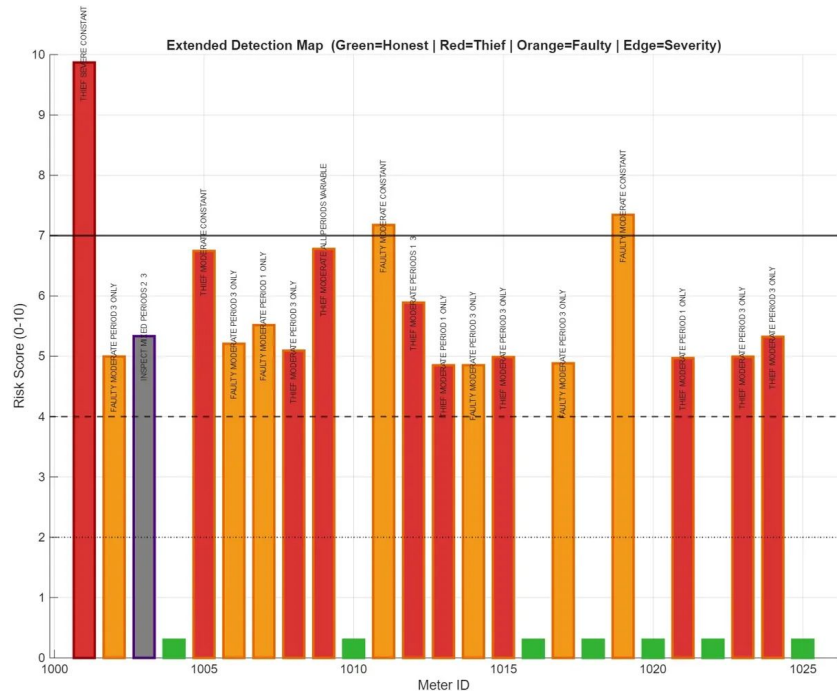
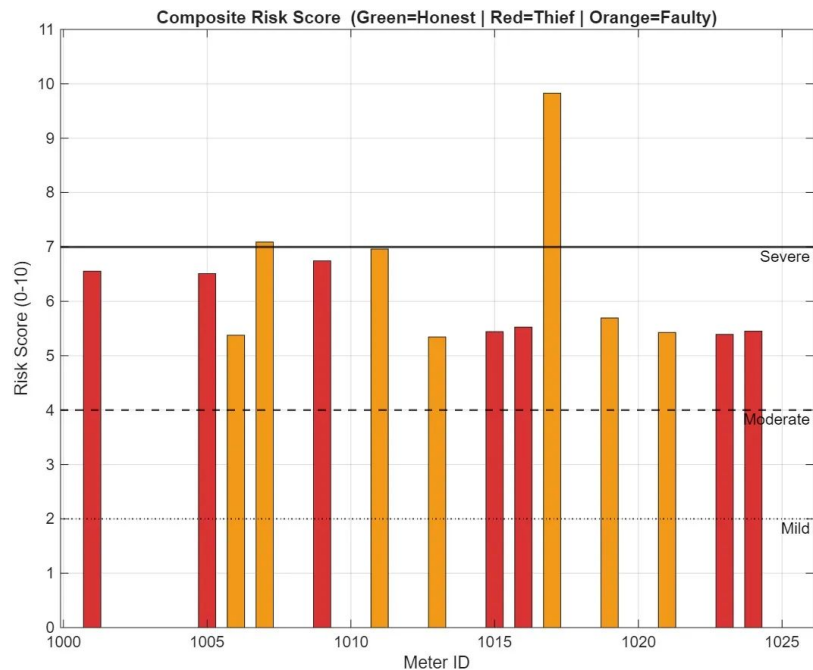
sults saved to cvir_adaptive_extended_results.csv

PRIORITY REPORT			
terID	Label	Risk	Loss(kWh) Action
01	THIEF_SEVERE_CONSTANT	9.9	26.912 URGENT - on-site inspection within 24 h. Affected: P1(00:00-08:00): under-reports 49% P2(08:00-16:00): under-reports 52% P3(16:00-24:00): under-reports 50%
19	FAULTY_MODERATE_CONSTANT	7.3	41.586 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): over-reads 50% P2(08:00-16:00): over-reads 73% P3(16:00-24:00): over-reads 50%
11	FAULTY_MODERATE_CONSTANT	7.2	36.388 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): over-reads 40% P2(08:00-16:00): over-reads 50% P3(16:00-24:00): over-reads 50%
09	THIEF_MODERATE_ALL_PERIODS_VARIABLE	6.8	24.091 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): under-reports 45% P2(08:00-16:00): under-reports 46% P3(16:00-24:00): under-reports 49%
05	THIEF_MODERATE_CONSTANT	6.7	23.056 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): under-reports 46% P2(08:00-16:00): under-reports 49% P3(16:00-24:00): under-reports 49%
12	THIEF_MODERATE_PERIODS_1_3	5.9	17.188 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): under-reports 24% P2(08:00-16:00): under-reports 6% P3(16:00-24:00): under-reports 6%
07	FAULTY_MODERATE_PERIOD_1_ONLY	5.5	26.283 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): over-reads 83% P2(08:00-16:00): over-reads 32% P3(16:00-24:00): over-reads 32%
03	INSPECT_MIXED_PERIODS_2_3	5.3	0.000 LOW - review records. Mixed anomaly - manual review required.
24	THIEF_MODERATE_PERIOD_3_ONLY	5.3	20.297 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): under-reports 16% P3(16:00-24:00): under-reports 44%
06	FAULTY_MODERATE_PERIOD_3_ONLY	5.2	16.723 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): over-reads 9% P2(08:00-16:00): over-reads 14% P3(16:00-24:00): over-reads 14%
08	THIEF_MODERATE_PERIOD_3_ONLY	5.1	13.166 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): under-reports 10% P3(16:00-24:00): under-reports 34%
02	FAULTY_MODERATE_PERIOD_3_ONLY	5.0	10.181 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): over-reads 10% P3(16:00-24:00): over-reads 74%
23	THIEF_MODERATE_PERIOD_3_ONLY	5.0	10.093 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): under-reports 12% P3(16:00-24:00): under-reports 26%
15	THIEF_MODERATE_PERIOD_3_ONLY	5.0	9.908 HIGH - schedule inspection within 1 week. Affected: P2(08:00-16:00): under-reports 10% P3(16:00-24:00): under-reports 24%
21	THIEF_MODERATE_PERIOD_3_ONLY	5.0	9.379 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): under-reports 32% P2(08:00-16:00): under-reports 7%
17	FAULTY_MODERATE_PERIOD_3_ONLY	4.9	6.705 HIGH - schedule inspection within 1 week. Affected: P3(16:00-24:00): over-reads 73%
14	FAULTY_MODERATE_PERIOD_3_ONLY	4.9	5.708 HIGH - schedule inspection within 1 week. Affected: P3(16:00-24:00): over-reads 37%
13	THIEF_MODERATE_PERIOD_1_ONLY	4.9	5.703 HIGH - schedule inspection within 1 week. Affected: P1(00:00-08:00): under-reports 25% P2(08:00-16:00): under-reports 6%

The model achieves 60% accuracy, with errors mainly being cautious false positives, ensuring real anomalies are not missed. It also provides precise insights by highlighting affected time periods and deviation levels (e.g., Meter 1001 under-reporting ~50% in P1–P3).



The Effective Anomaly Chart clearly validates the adaptive period approach, showing period-wise deviations against the threshold (0.249). It enables precise detection of constant anomalies (e.g., Meter 1001) and time-specific issues (e.g., Meter 1015 in P3), which was not possible with fixed 2-period analysis.



While severity grading (Modification 1) remains unchanged, risk scores are now more accurate due to finer time-window analysis, reducing averaging effects.

The Extended Detection Map consolidates class, severity, risk, and period patterns, clearly highlighting high-risk cases (e.g., Meter 1001) while safely handling ambiguous ones (e.g., Meter 1003 as INSPECT).

Conclusion

This project develops a robust, data-driven framework for detecting non-technical losses (NTL) and meter faults in AMI systems. It highlights the limitations of basic regression and introduces the CVLR-ETDM approach for capturing time-varying anomalies using temporal segmentation and severity analysis.

The system effectively distinguishes between energy theft and meter degradation while generating an automated Priority Report for actionable insights. This reduces manual inspections, lowers financial losses, and improves overall grid reliability, with future scope for integration with real-time machine learning for predictive detection.

Pseudo code of CVLR ETDM

Load Data

INPUT:

Smart meter data (MeterID, TimeCode, Usage)

Collector data (TimeCode, Total Supply)

EXTRACT:

Unique meter IDs $\rightarrow N$

Unique time slots $\rightarrow T$

Build Consumption Matrix

Initialize matrix P of size $(T \times N)$

FOR each meter $i = 1$ to N :

Extract usage values for meter i

Map time slots to correct indices

Fill column i of P with usage values

END

Compute Energy Mismatch

Sort collector data by time

c = total supplied energy

$y(t) = c(t) - \text{sum of all meter consumptions at time } t$

Split into Peak and Off-Peak

Check Data Sufficiency

IF $T \leq 2 \times N$:

STOP (Not enough data)

Construct Regression System

Build matrix A:

$A = \begin{bmatrix} P_{\text{off}} & 0 \\ P_{\text{peak}} & P_{\text{peak}} \end{bmatrix}$

Stack output:

$y_{\text{stack}} = \begin{bmatrix} y_{\text{off}} \\ y_{\text{peak}} \end{bmatrix}$

Solve Using Ridge Regression

Compute: $\theta = (A^T A + \lambda I)^{-1} A^T y_{\text{stack}}$

Split: α (alpha) = first N values of θ ; β (beta) = next N values

Compute: $\alpha + \beta$ (peak anomaly)

Statistical Testing

For each meter i : Compute standard error (SE), t-statistic, p-value

Set Thresholds

sig level and zero_tol

Classify logics

Estimate Multiplier FOR each meter i :

IF anomaly detected: $m = 1 / (1 + \text{anomaly coefficient})$

ELSE: $m = 1$

Compute: Accuracy = (Correct Predictions / Total Meters) $\times 100$

Save results and plots

Pseudo code of modifications

Load data

BUILD CONSUMPTION MATRIX (P) : $T \times N$ matrix of zeros

$y \leftarrow c - \text{row_sum}(P)$

AUTO-SELECT NUMBER OF PERIODS

Choose largest $k \in [\text{MIN_PERIODS}, \text{MAX_PERIODS}]$ such that: $T - (N \times k) \geq \text{MIN_DF}$

Set: $\text{NPERIODS} = k$, $\text{nParams} = N \times k$, $\text{df} = T - \text{nParams}$

If $k < \text{MAX_PERIODS}$: estimate additional data required

DEFINE TIME PERIODS

$\text{slotsPerPeriod} \leftarrow \text{floor}(48 / \text{NPERIODS})$

FOR each period k : DEFINE start_slot and end_slot ; ENSURE last period ends at slot 47; CREATE human-readable names for periods

BUILD DESIGN MATRIX (A)

INITIALIZE A as $T \times \text{nParams}$

FOR each period k : $\text{rows_k} \leftarrow$ rows belonging to period k

Alpha block (always active): $A(\text{rows_k}, 1:N) \leftarrow P(\text{rows_k}, :)$

Beta block (only for $k \geq 2$)

IF $k \geq 2$:

$\text{column_start} \leftarrow (k-1)*N + 1$

$\text{column_end} \leftarrow k*N$

$A(\text{rows_k}, \text{column_start}:\text{column_end}) \leftarrow P(\text{rows_k}, :)$

RIDGE REGRESSION

$A^T A \leftarrow A^T \times A$

GENERATE λ candidates (log scale)

FOR each λ :

$H \leftarrow A \times \text{inverse}(A^T A + \lambda I) \times A^T$

$\hat{y} \leftarrow H \times y$

$\text{residual} \leftarrow y - \hat{y}$

COMPUTE GCV score: $\text{gcv} \leftarrow \text{mean}(\text{residual}^2) / (1 - \text{trace}(H)/T)^2$

SELECT λ with minimum GCV

SOLVE FOR PARAMETERS t_{stat} , p values, SE

EXTRACT PARAMETERS

$\alpha \leftarrow$ first N values of θ

IF $\text{NPERIODS} > 1$: $\beta \leftarrow$ reshape remaining values into matrix ($N \times (\text{NPERIODS}-1)$)

ELSE: $\beta \leftarrow$ empty

COMPUTE EFFECTIVE ANOMALY

FOR each consumer i : $\text{eff}(i, 1) \leftarrow \alpha(i)$

FOR each period $k \geq 2$: $\text{eff}(i, k) \leftarrow \alpha(i) + \beta(i, k-1)$

DEFINE THRESHOLDS

$\text{ZERO_TOL} \leftarrow \sigma_y / (\text{mean_p} \times \sqrt{N})$

CLASSIFICATION LOOP (FOR EACH CONSUMER)

Validation and then saved results